Suggested problems §20.1: 3-9 odd

Suggested problems §20.2: 1-11 odd

Problem 1: Compute the curl $\nabla \times \vec{F}$ for each of the following vector fields.

(a) $\vec{F}(x, y, z) = 3x\vec{i} - 5z\vec{j} + y\vec{k}$

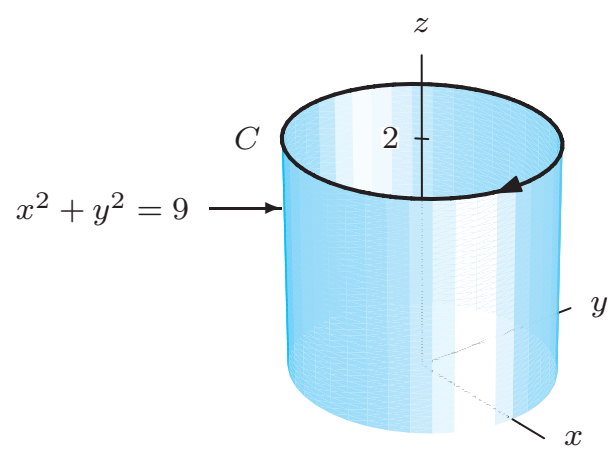
$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x & -5z & y \end{vmatrix} = \langle 6, 0, 0 \rangle$$

(b) $\vec{F}(x, y, z) = \langle e^x, z\cos y, e^{y^2} \rangle$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x & z\cos y & e^{y^2} \end{vmatrix} = \langle 2ye^{y^2} - \cos y, 0, 0 \rangle$$

Problem 2: The figure to the right shows an open cylindrical can S standing on the xy-plane. (S has a bottom and sides, but no top.) If S is oriented outward and downward, compute

$$\int_S \nabla \times \langle -y, x, z \rangle \cdot d\vec{A}.$$



We use Stokes' Theorem, where C is the boundary curve. The curve is parameterized by

$$\vec{r}(t) = \langle 3\sin t, 3\cos t, 2 \rangle$$

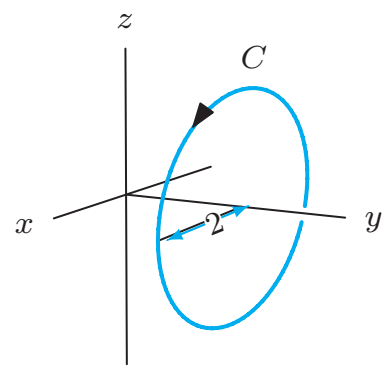
for $t \in [0, 2\pi]$. Then

$$\begin{aligned} \int_S (\nabla \times \langle -y, x, z \rangle) \cdot d\vec{A} &= \oint_C \langle -y, x, z \rangle \cdot d\vec{r} \\ &= \int_0^{2\pi} \langle -3\cos t, 3\sin t, 2 \rangle \cdot \langle 3\cos t, -3\sin t, 0 \rangle \, dt \\ &= -9 \int_0^{2\pi} dt \\ &= -18\pi. \end{aligned}$$

Problem 3: Compute

$$\oint_C (-z\vec{i} + y\vec{j} + x\vec{k}) \cdot d\vec{r}$$

where C is the circle of radius 2 shown to the right.



We use Stokes' Theorem, where S is the solid disk bounded by C. We first compute the curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -z & y & x \end{vmatrix} = \langle 0, -2, 0 \rangle.$$

Stokes' Theorem then gives us

$$\oint_C (-z\vec{i} + y\vec{j} + x\vec{k}) \cdot d\vec{r} = \int_S \langle 0, -2, 0 \rangle \cdot d\vec{A},$$

where S is oriented away from the origin. But since S is flat and $\langle 0, -2, 0 \rangle$ is constant, we have

$$\int_S \langle 0, -2, 0 \rangle \cdot d\vec{A} = \langle 0, -2, 0 \rangle \cdot \vec{A} = \langle 0, -2, 0 \rangle \cdot \langle 0, 4\pi, 0, \rangle = -8\pi,$$

where $\vec{A} = \langle 0, 4\pi, 0 \rangle$ is the area vector of S.

Problem 4: Compute

$$\oint_C \langle x^2, y^2, z^2 \rangle \cdot d\vec{r}$$

where C is the unit circle in the xz-plane, oriented counterclockwise when viewed from the positive y-axis.

Let S be the solid disk bounded by C. We first compute the curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & y^2 & z^2 \end{vmatrix} = \langle 0, 0, 0 \rangle.$$

Thus if we use Stokes' Theorem, where S is oriented toward the negative y-axis, then we have

$$\oint_C \langle x^2, y^2, z^2 \rangle \cdot d\vec{r} = \int_S (\nabla \times \vec{F}) \cdot d\vec{A} = 0.$$

Problem 5: Compute

$$\oint_C (-y\vec{i} + x\vec{j} + z\vec{k}) \cdot d\vec{r}$$

where C is the circle of radius 2 in the plane $x + y + z = 3$, centered at $(1, 1, 1)$ and oriented clockwise when viewed from the origin.

Let S be the solid disk bounded by C. We first compute the curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & z \end{vmatrix} = \langle 0, 0, 2 \rangle.$$

Stokes' Theorem then gives us

$$\oint_C (-y\vec{i} + x\vec{j} + z\vec{k}) \cdot d\vec{r} = \int_S \langle 0, 0, 2 \rangle \cdot d\vec{A},$$

where S is oriented away from the origin. But since S is flat and $\langle 0, 0, 2 \rangle$ is constant, we have

$$\int_S \langle 0, 0, 2 \rangle \cdot d\vec{A} = \langle 0, 0, 2 \rangle \cdot \vec{A} = \langle 0, 0, 2 \rangle \cdot \frac{4\pi \langle 1, 1, 1 \rangle}{\sqrt{3}} = \frac{8\pi}{3},$$

where

$$\vec{A} = \frac{4\pi \langle 1, 1, 1 \rangle}{\sqrt{3}}$$

is the area vector of S.